

Treat the patient with TB¹ and HIV

- If already on dolutegravir-based ART regimen and starting TB treatment: increase **dolutegravir** (DTG) dose to 50mg 12 hourly². Continue this dose until 2 weeks after TB treatment completed.
- If already on TB treatment and starting ART (patient has never been on ART): consider **TEE (TDF + FTC + EFV)**. Switch to DTG-based regimen 2 weeks after TB treatment complete.
- Avoid atazanavir with rifampicin. If already on atazanavir, discuss with HIV expert, infectious disease specialist or HIV hotline ☎ 178.
- If on lopinavir/ritonavir, increase **lopinavir/ritonavir** dose gradually:
 - After 1 week of TB treatment, increase **lopinavir/ritonavir** to 600/150mg (3 tablets) 12 hourly for 1 week.
 - Then increase **lopinavir/ritonavir** to 800/200mg (4 tablets) 12 hourly. Continue this dose until 2 weeks after TB treatment completed.
 - Monitor for liver problem (jaundice, abdominal pain, vomiting) and check ALT monthly. If symptomatic with ALT > 120, or asymptomatic with ALT ≥ 200, refer.
 - Aim to switch to a dolutegravir-based regimen 2 weeks after TB treatment completed.

Look for and manage TB treatment side effects

Side effect	Likely cause	Management
Jaundice	Most TB medications	Stop all medications and refer to hospital same day.
Nausea, vomiting, abdominal pain	Most TB medications	• Check ALT and review result within 24 hours: - If ALT > 120, stop all medications and refer to hospital same day. - If ALT 50-120, assess for possible causes, consider interrupting treatment and repeat ALT within 1 week. If unsure, discuss or refer. • If nausea/vomiting: advise to take treatment at night. If significant nausea/vomiting, give metoclopramide 10mg 30 minutes before TB medication.
Skin rash/itch	Most TB medications	Assess and manage ☎ 67.
Seizures	Levofloxacin, isoniazid	Manage seizure ☎ 19 and refer to hospital same day.
Psychosis	Levofloxacin, isoniazid	Manage psychosis ☎ 85 and discuss/refer to hospital same day.
Change in vision	Ethambutol	Stop ethambutol and refer to eye specialist same day.
Joint pain	Pyrazinamide, levofloxacin	Give ibuprofen 400mg 8 hourly as needed with food for up to 5 days (avoid if peptic ulcer, asthma, hypertension, heart failure or kidney disease).
Orange urine	Rifampicin	Reassure this is normal while taking rifampicin.
Pain/numbness of feet	Isoniazid	Peripheral neuropathy likely ☎ 66.

Manage the patient with DS-TB and a positive week 7 smear

- Look for explanation for result: if poor adherence, give enhanced adherence support ☎ 173, alcohol/drug use ☎ 142, stress ☎ 86 or treatment side effects (see above).
- Send 1 sputum for culture and DST and continue intensive phase treatment for a month.
- Follow up culture and DST result every 1-2 weeks until available and review:

Sensitive to rifampicin and INH		Resistant to INH only	Resistant to rifampicin
• Change to continuation phase ☎ 96 and monitor clinically. • Repeat TB microscopy (smear) at week 11:		Diagnose INH mono-resistant TB • Start treatment same day →95. • Register patient as INH mono-resistant TB.	Diagnose RR-TB • Stop DS-TB treatment: - If resistant to rifampicin only, give outcome of "rifampicin resistant TB" in patient folder. - If resistant to rifampicin and INH, give outcome of "multidrug-resistant TB" in patient folder. • Give routine RR-TB care and start treatment same day →99. • Register in drug-resistant TB register. • If RR-TB care not available, refer to drug-resistant TB initiation facility.
Smear positive	Smear negative		
• Assess clinically: - If patient improving, change to continuation phase and continue treatment for a total of 6 months. - If patient deteriorating, refer.	• Continue treatment and monitor clinically. • Continue treatment for a total of 6 months.		

¹This includes drug-sensitive TB (DS-TB) and INH-mono-resistant TB. ²If on fixed dose combination, tenofovir/lamivudine/dolutegravir (TLD): continue this and add **dolutegravir** 50mg 12 hours after TLD dose.